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IMAGE PICKUP APPARATUS THAT CONTROLS DISPLAY OF RETURN VIDEO, CONTROL METHOD THEREFOR, AND STORAGE MEDIUM

Abstract

An image pickup apparatus that can switch a method for displaying a return video in accordance with the need to display the return video, thereby reducing the inhibition caused by displaying a return video when displaying a self-shot video. The image pickup apparatus connected to an external apparatus receives a tally signal and a return video, and switches a method for displaying the return video. The method for displaying the return video is switched depending on a state of the tally signal.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an image pickup apparatus, a control method therefor, and a storage medium, and more particularly, to an image pickup apparatus that controls the display of a return video, a control method therefor, and a storage medium.

Description of the Related Art

[0002] Conventionally, there exist a plurality of image pickup apparatuses constituting a photographic system, some are configured to display not only a self-shot video being shot by themselves but also a currently broadcasted/distributed video that has been received from a switcher as a return video.

[0003] To display such a return video, more specifically, to switch between the displays of a self-shot video and a return video, various techniques are utilized. For example, Japanese Laid-Open Patent Publication (kokai) No. 2011-223541 proposes a technique in which a user operates an external remote control to manually switch between the displayed and non-displayed states of a return video on a display unit of an image pickup apparatus.

[0004] In the technique proposed in Japanese Laid-Open Patent Publication (kokai) No. 2011-223541, however, the return video is displayed on the display unit of the image pickup apparatus in response to the user's operation on the remote control, regardless of whether the need to display the return video is great or small. Therefore, there is a case where the display of a self-shot video is inhibited by the display of a return video that is less necessary. If a return video is displayed on a self-shot video in a picture-in-picture (PinP) mode, for example, it may be impossible to view the area of the self-shot video over which the return video is displayed.

SUMMARY OF THE INVENTION

[0005] The present invention provides an image pickup apparatus that can switch a method for displaying a return video in accordance with the need to display the return video, thereby reducing the inhibition caused by displaying a return video when displaying a self-shot video, a control method therefor and a storage medium.

[0006] Accordingly, the present invention provides an image pickup apparatus connected to an external apparatus, comprising at least one memory and at least one processor which function as a first receiving unit configured to receive a tally signal, a second receiving unit configured to receive a return video, and a control unit configured to switch a method for displaying the return video, wherein the control unit switches the method for displaying the return video, depending on a state of the tally signal in the first receiving unit.

[0007] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a block diagram showing a hardware configuration of a digital video camera, which is an image pickup apparatus according to an embodiment of the present invention.

[0009] FIG. 2 is a view showing an example of switching a method for displaying a return video in a case of a setting in which the return video is displayed in a PinP mode with respect to a self-shot video.

[0010] FIG. 3 is a diagram showing an example of switching a method for displaying a return video in a case of a setting in which the return video is displayed in a mixing mode with respect to

a self-shot video.

[0011] FIG. 4 is a diagram showing an example of switching a method for displaying a return video in a case of a setting in which the return video and a self-shot video are displayed in a screen separation mode.

[0012] FIG. 5 is a flowchart showing an example of a display switching control process for a return video.

DESCRIPTION OF THE EMBODIMENTS

[0013] The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.

[0014] Some embodiments of the present invention will be described below in detail with reference to the drawings. Each of the embodiments to be described below is merely an example of configurations that can implement the present invention. The following embodiments can be appropriately modified or varied in accordance with the configuration of an apparatus to which the present invention is to be applied and various conditions. Furthermore, not all conceivable combinations of elements included in the following embodiments are essential to implement the present invention, and some of the elements can be appropriately omitted. Therefore, the scope of the present invention is not limited by the configurations described in the following embodiments. In addition, a plurality of the configurations described in the embodiments can be combined together unless they contradict with each other.

[0015] FIG. 1 is a block diagram showing a hardware configuration of a digital video camera **10**, which is an image pickup apparatus according to an embodiment of the present invention.

[0016] As shown in FIG. 1, the digital video camera **10** is communicably connected to a recording medium **120**, a display unit **121**, and a switcher **128** that are all disposed outside.

[0017] As shown in FIG. 1, the digital video camera **10** includes a housing **100** that contains a plurality of elements. The elements contained in the digital video camera **10** include a video processing unit **103**, a codec **104**, a video output unit **105**, a memory control unit **107**, an OSD rendering unit **118**, an I/F **119**, and a system control unit **109**. The elements contained in the digital video camera **10** also include a tally signal receiving unit **122**, a return video receiving unit **124**, and a return video transmitting unit **126**. Those elements are interconnected via a bus.

[0018] The elements contained in the digital video camera **10** further include a lens barrel **101**, an image pickup unit **102**, a mode switch **110**, and an operation unit **111**. The elements contained in the digital video camera **10** also include a power switch **112**, a nonvolatile memory **113**, a power control unit **114**, a power supply unit **115**, a system memory **116**, and a system timer **117**.

[0019] The system control unit **109** is connected to the lens barrel **101**, the image pickup unit **102**, the mode switch **110**, the operation unit **111**, the power switch **112**, the nonvolatile memory **113**, the power control unit **114**, the system memory **116**, and the system timer **117**.

[0020] The lens barrel **101** has image pickup lenses, including a focus lens and a stabilizing lens, and a diaphragm mechanism. The lens barrel **101** may be integrated with the housing **100** or may be detachable from the housing **100**.

[0021] The image pickup unit **102** includes: an image pickup sensor that converts an optical image collected and formed by the image pickup lenses disposed inside the lens barrel **101** into an analog electrical signal; and an A/D converter that converts the analog electrical signal into a digital signal (video data).

[0022] The video processing unit **103** subjects the video data from the image pickup unit **102** or the memory control unit **107** to predetermined image processes, such as a resizing process, a trimming process, a color converting process, and a barrel distortion correcting process and then generates a VRAM in a memory **108** via the memory control unit **107**. In addition, the video processing unit **103** subjects the video data to a predetermined arithmetic process and supplies the arithmetic result to the system control unit **109**. This calculation result is used for various control processes, such as exposure control, ranging control, and stabilizing control. Based on the result of subject detection

which has been calculated by the video processing unit **103**, for example, the video processing unit **103** performs an autofocus (AF) process, an automatic exposure (AE) process, and a stabilizing process. In addition, based on the arithmetic result on the video data, the video processing unit **103** can perform an automatic white balance (AWB) process.

[0023] The codec **104** encodes the VRAM generated by the video processing unit **103** in conformity with a predetermined moving image compression scheme (such as MPEG-2 or H.264). In addition, the codec **104** decodes encoded video data supplied from the memory control unit **107** to generate a new VRAM. This new VRAM is subjected to a predetermined image process by the video processing unit **103** and then supplied to the memory control unit **107**.

[0024] The video output unit **105** superimposes a plurality of VRAMs that has been read from the memory **108** via the memory control unit **107**, thereby generating a video signal. In this case, the video output unit **105** can add metadata specified by the system control unit **109** to this video signal. It should be noted that the video output unit **105** can independently generate individual video signals for a plurality of display parts **121** and output the video signals via a display control unit **106**.

[0025] The display unit **121** is a display apparatus connected to digital video camera **10** and displays an input video signal. Alternatively, the display unit **121** may be provided in the digital video camera **10**. The display unit **121** can have a plurality of display apparatuses. The display unit **121** may be formed of a display member, such as a liquid crystal panel or an organic EL panel. Alternatively, the display unit **121** may be an external recording apparatus for recording a video signal, which is disposed outside the digital video camera **10**.

[0026] The display control unit **106** establishes the connection to the display unit **121** and outputs the video signal that has been output from the video output unit **105**, to the display unit **121**. In short, a self-shot video being shot by the digital video camera **10** (or a live video being picked up by the image pickup unit **102** and being subjected to the image process by video processing unit **103**) is output from the video output unit **105** and displayed on the display unit **121**. The digital video camera **10** and the display unit **121** are connectable together in conformity with the serial digital interface (SDI) standard, for example. It should be noted that the digital video camera **10** and the display unit **121** may be connected in a wired or wireless manner with another technique. For example, the digital video camera **10** and the display unit **121** may be connected together by an HDMI (registered trademark) cable, so that a video signal and signal specification information may be transmitted from one of the digital video camera **10** and the display unit **121** to the other. In addition, the digital video camera **10** and the display unit **121** may be connected via a wireless LAN.

[0027] The memory control unit **107** mediates an access request from each individual unit to the memory **108**.

[0028] The memory **108** stores VRAMs to be handled by the video processing unit **103**, the codec **104**, the video output unit **105**, and the OSD rendering unit **118**. In addition, the memory **108** can temporarily store the encoded video data that has been output from the codec **104** and encoded video data that has been read from the recording medium **120**. The memory **108** has an enough recording capacity to store a moving image and sound for a predetermined time.

[0029] The system control unit **109** is a control unit that integrally controls the whole of the digital video camera **10** and is formed of an arithmetic processor, such as a central processing unit (CPU). Alternatively, the system control unit **109** may include a plurality of CPU cores. The CPU cores can independently process tasks described in a program, which will be described later.

[0030] The nonvolatile memory **113** is an electrically recordable and erasable recording medium. The nonvolatile memory **113** records control information, such as programs and parameters, to be used by the system control unit **109**. These programs include a program for performing a process shown in a flowchart, which will be described later.

[0031] The system memory **116** is a volatile recording medium and is formed of a random access

memory (RAM), for example. It should be noted that the system memory **116** may be used commonly with the memory **108**. In this case, the access to the memory **108** serving as the system memory **116** is mediated by the memory control unit **107**. Thus, a small-capacity, fast-accessible memory (such as a cache memory) may be provided and directly connected to the system control unit **109**.

[0032] The system control unit **109** (control unit) deploys, in the system memory **116**, programs and operational variables and constants stored in the nonvolatile memory **113** and executes the programs, thereby controlling each individual unit in the digital video camera **10**. As a result, the system control unit **109** realizes individual operations of the present embodiment, such as a display switching control process for a return video, which will be described later with reference to FIG. 5. The system control unit **109** can control display operations to be performed by the memory **108**, the OSD rendering unit **118**, and the video output unit **105**.

[0033] The mode switch **110** is a switch used to select in which mode the digital video camera **10** is to operate. The operation mode (such as a camera mode or a reproduction mode) designated by the position of the switch is reported to the system control unit **109**.

[0034] The operation unit **111** is an operation unit used to enter various instructions in the system control unit **109**. The operation unit **111** includes, for example, the following elements: [0035] a menu button through which a menu screen for performing various settings is to be displayed on the display unit **121**; [0036] a cancel button; [0037] arrow keys (up arrow key, down arrow key, left arrow key, and right arrow key); [0038] A SET button; [0039] a function switching button (AF/MF switching button); [0040] a REC button through which the recording of a moving image is to be started or stopped; and [0041] an assignment button through which any function can be assigned to the menu setting;

[0042] Using the menu screen displayed on the display unit **121**, the arrow keys, and the SET button, a user can intuitively give various setting instructions. It should be noted that the user may operate the REC button to give the recording start/stop instruction for recording a video signal on the display unit **121** (external recording apparatus).

[0043] The power switch **112** is a push button used to switch between a power-on state and a power-off state.

[0044] The power control unit **114** includes a battery detection circuit, a DC-DC converter, and a switching circuit that switches between blocks to be energized and detects whether a battery is attached, which type of the battery is attached, and how much the battery level remains. The power control unit **114** controls the DC-DC converter based on the above detection result and an instruction from the system control unit **109** and applies a voltage required for the operation to each unit (e.g., the recording medium **120**) over a necessary period.

[0045] The power supply unit **115** is formed of at least one of a primary battery (alkaline battery, lithium battery, etc.), a secondary battery (NiCd battery, NiMH battery, Li battery, etc.), and an AC adapter.

[0046] The system timer **117** is a timing generating unit used for various types of control and is also a clocking unit that reads a time indicated by a built-in clock (not shown). The system control unit **109** controls the operation of each unit based on a timing generated by the system timer **117**.

[0047] The OSD rendering unit **118** provides an on-screen display (OSD) function by which multiple pieces of setting information are superimposed on video data. More specifically, the OSD rendering unit **118** renders character strings and icons representing the state and settings of the digital video camera **10** as well as various display frames and markers (i.e., the OSD) to the VRAMs stored in the memory **108**. Display information, such as the character strings and the icons, used for the above on-screen display function is stored in the nonvolatile memory **113** and, as necessary, is read by the OSD rendering unit **118**.

[0048] The I/F **119** is a connection interface that enables the recording medium **120** (external recording apparatus) to be connected to the digital video camera **10**.

[0049] The recording medium **120** is a recording medium on which encoded video data and data associated with the video data are to be stored as image files. The encoded video data stored on the memory **108** is supplied to the recording medium **120** via the I/F **119** and is recorded thereon. The encoded video data and the associated data that have been stored on the recording medium **120** are read at the I/F **119** and transferred to the memory **108**. The recording medium **120** may be a medium (memory card, HDD, other discs, etc.) to be detachably attached to the digital video camera **10** or a medium (flash memory, HDD, etc.) disposed inside the digital video camera **10**.

[0050] The switcher **128** (external apparatus) is a device connected to the digital video camera **10** and includes: a tally signal output unit **123** that transmits a tally signal; a return video output unit **125** that transmits a return video; and a return video input unit **127** that receives the return video.

[0051] The tally signal receiving unit **122** (first receiving unit) receives the tally signal from the tally signal output unit **123** in the switcher **128** disposed outside. The reception of the tally signal in this case is reported to the system control unit **109**. In response to this, the system control unit **109** causes the OSD rendering unit **118** to perform drawing in accordance with the content of the received tally signal. As a result, the OSD rendering unit **118** draws, for example, an icon that indicates the reception of the tally signal and/or a frame (image frame) for changing the color of the periphery of the image displayed on the display unit **121**.

[0052] The return video receiving unit **124** (second receiving unit) receives the return video from the return video output unit **125** in the switcher **128** disposed outside. The video data received in this case is stored on the memory **108**, processed by the video output unit **105** under the control of the system control unit **109**, and displayed on the display unit **121**.

[0053] The return video transmitting unit **126** transmits the return video to the return video input unit **127** in the switcher **128** disposed outside.

[0054] FIG. **2** is a view showing an example of switching a method for displaying the return video in a case of a setting (first setting) in which the return video is displayed in a picture-in-picture (PinP) mode with respect to a self-shot video.

[0055] Hereinafter, the state “tally signal: none” (second state) refers to a state where the tally signal receiving unit **122** is not receiving the tally signal. The state “tally signal: PROGRAM” (first state) refers to a state where the tally signal receiving unit **122** is receiving a tally signal of PROGRAM. Herein, the tally signal of PROGRAM refers to a tally signal indicating that the operator of the switcher **128** selects the video being shot by the digital video camera **10** as the video (return video) to be currently broadcasted. The state “tally signal PREVIEW” (third state) refers to a state where the tally signal receiving unit **122** is receiving a tally signal of PREVIEW. Herein, the tally signal of PREVIEW refers to a tally signal indicating that the operator of the switcher **128** selects the video being shot by the digital video camera **10** as the video that will be broadcasted/distributed next.

[0056] A screen **200a** in FIG. **2** corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is in the state “tally signal: none”, where a return video **201** is displayed in the PinP mode simultaneously with a self-shot video.

[0057] A screen **200b** in FIG. **2** corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is switched to the state “tally signal: PROGRAM” during the displaying of the screen **200a**. In this case, the return video **201** that has been displayed in the PinP mode on the self-shot video is switched to the non-displayed state. Although not shown in FIG. **2**, when the tally signal receiving unit **122** is in the state “tally signal: PROGRAM”, a red frame is optionally displayed around the screen **200b**.

[0058] When the tally signal receiving unit **122** is returned from the state “tally signal: PROGRAM” to the state “tally signal: none”, the screen **200b** is switched back to the screen **200a** on which the return video **201** is displayed in the PinP mode.

[0059] This functionality is employed in consideration of the following fact: when the tally signal receiving unit **122** is in the state “tally signal: PROGRAM”, the return video is a video based on

the self-shot video, in which case it is generally less necessary to display the return video on the display unit **121**.

[0060] Although not shown in FIG. 2, when the tally signal receiving unit **122** is switched to the state “tally signal: PREVIEW”, the screen **200b** is switched to a screen **200c** on which the return video **201** is displayed so that the visibility of the self-shot video becomes higher than that on the screen **200b**. Examples of a method for increasing the visibility of the self-shot video include: a method for downsizing the display (PinP display) of the return video **201**; and a method for increasing the transmittance of the return video displayed in the PinP mode. Although not shown in FIG. 2, when the tally signal receiving unit **122** is in the state “tally signal: PREVIEW”, a green frame may be displayed around the screen **200c**.

[0061] The reduction proportion and transmittance of the return video displayed in the PinP mode may be set at the initial setup of the digital video camera **10**, depending on the state of the tally signal receiving unit **122**. Alternatively, the reduction proportion and transmittance thereof may be set changeable through a user's operation on the operation unit **111**.

[0062] FIG. 3 is a view showing an example of switching a method for displaying the return video in a case of a setting (second setting) in which the return video is displayed in a mixing (translucent superimposing) mode with respect to a self-shot video.

[0063] A screen **300a** in FIG. 3 corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is in the state “tally signal: none”, where a return video **202** is displayed in the mixing mode simultaneously with a self-shot video.

[0064] A screen **300b** in FIG. 3 corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is switched to the state “tally signal: PROGRAM” during the displaying of the screen **300a**. In this state, the return video **202** that has been displayed on the self-shot video in the mixing mode is switched to the non-displayed state.

[0065] When the tally signal receiving unit **122** is returned from the state “tally signal: PROGRAM” to the state “tally signal: none”, the screen **300b** is switched back to the screen **300a** on which the return video **202** is displayed in the mixing mode.

[0066] This functionality is employed in consideration of the following fact: when the tally signal receiving unit **122** is in the state “tally signal: PROGRAM”, the return video is a video based on the self-shot video, in which case it is generally less necessary to display the return video on the display unit **121**.

[0067] Although not shown in FIG. 3, when the tally signal receiving unit **122** is switched to the state “tally signal: PREVIEW”, the screen **300b** is switched to a screen **300c** on which the return video **202** is displayed so that the visibility of the self-shot video becomes higher than that on the screen **300b**. Examples of a method for increasing the visibility of the self-shot video include a method for increasing the transmittance of the return video **202** displayed in the mixing mode.

[0068] The transmittance of the return video displayed in the mixing mode may be set at the initial setup of the digital video camera **10**, depending on the state of the tally signal receiving unit **122**. Alternatively, the transmittance thereof may be set changeable through a user's operation on the operation unit **111**.

[0069] FIG. 4 is a diagram showing an example of switching a method for displaying the return video in a case of a setting (third setting) in which the return video and the self-shot video are displayed in a screen separation mode.

[0070] A screen **400a** in FIG. 4 corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is in the state “tally signal: none”, where, a return video **203** and a self-shot video **401** are displayed in the screen separation mode.

[0071] A screen **400b** in FIG. 4 corresponds to a screen displayed on the display unit **121** when the tally signal receiving unit **122** is switched to the state “tally signal: PROGRAM” during the displaying of the screen **400a**. In this state, the return video **203** which has been displayed in the screen separation mode is switched to the non-displayed state, whereas the self-shot video **401**

which has been displayed in the screen separation mode is displayed in the full-screen mode.

[0072] When the tally signal receiving unit **122** is returned from the state “tally signal: PROGRAM” to the state “tally signal: none”, the screen **400b** is switched back to the screen **400a** on which the return video **203** is displayed in the screen separation mode.

[0073] This functionality is employed in consideration of the following fact: when the tally signal receiving unit **122** is in the state “tally signal: PROGRAM”, the return video is a video based on the self-shot video, in which case it is generally less necessary to display the return video on the display unit **121**.

[0074] Although not shown the screen in FIG. **4**, when the tally signal receiving unit **122** is switched to the state “tally signal: PREVIEW”, the screen **400b** is switched to a screen **400c** on which the return video **203** is displayed so that the visibility of the self-shot video becomes higher than that on the screen **400b**.

[0075] Examples of a method for increasing the visibility of the self-shot video include: a method for displaying, in the screen separation mode, the return video **203** with a decreased size and the self-shot video with an increased size.

[0076] The display sizes of the return video **201** and the self-shot video may be set at the initial setup of the digital video camera **10**, depending on the state of the tally signal receiving unit **122**. Alternatively, the display sizes thereof may be set changeable through a user's operation on the operation unit **111**.

[0077] It should be noted that any of the methods shown in FIGS. **2** to **4** may be utilized as long as a self-shot video and a return video are displayed simultaneously on the display unit **121**. Alternatively, any other method that does not limit the present invention may be utilized.

Furthermore, one of the methods for simultaneously displaying a self-shot video and a return video on the display unit **121** may be set in an initial setting stage of the digital video camera **10**, or the methods may be set changeable in response to a user's operation on the operation unit **111**.

[0078] FIG. **5** is a flowchart showing an example of a display switching control process for a return video.

[0079] In step **S500**, the system control unit **109** determines whether the return video receiving unit **124** is receiving the return video. When the return video receiving unit **124** is receiving the return video (YES in step **S500**), the process proceeds to step **S501**. When the return video receiving unit **124** is not receiving the return video (NO in step **S500**), the process proceeds to step **S507**. In step **S501**, the system control unit **109** determines whether the display unit **121** is set to display the return video. When the display unit **121** is set to display the return video (YES in step **S501**), the process proceeds to step **S502**. When the display unit **121** is not set to display the return video (NO in step **S501**), the process proceeds to step **S507**.

[0080] In step **S502**, the system control unit **109** determines whether the tally signal receiving unit **122** is receiving the tally signal of PROGRAM (whether the tally signal receiving unit **122** is in the state “tally signal: PROGRAM”). When the tally signal receiving unit **122** is receiving the tally signal of PROGRAM (YES in step **S502**), the process proceeds to step **S506**. When the tally signal receiving unit **122** is not receiving the tally signal of PROGRAM (NO in step **S502**), the process proceeds to step **S503**.

[0081] In step **S503**, the system control unit **109** determines whether the tally signal receiving unit **122** is receiving the tally signal of PREVIEW (whether the tally signal receiving unit **122** is in the state “tally signal: PREVIEW”). When the tally signal receiving unit **122** is receiving the tally signal of PREVIEW (YES in step **S503**), the process proceeds to step **S505**. When the tally signal receiving unit **122** is not receiving the tally signal of PREVIEW (NO in step **S503**), the process proceeds to step **S504**.

[0082] In step **S504**, the system control unit **109** determines that the tally signal receiving unit **122** is in the state “tally signal: none” and controls the display unit **121** in such a way that the display unit **121** simultaneously displays both of the self-shot video and the return video. The process then

returns to step S500.

[0083] In step S505, the system control unit **109** determines that the tally signal receiving unit **122** is in the state “tally signal: PREVIEW” and controls the display unit in such a way that the display unit displays both of the self-shot video and the return video. In this case, the system control unit **109** controls the display unit in such a way that the display unit displays the return video with the visibility of the self-shot video being higher than that in step S504. The process then returns to step S500.

[0084] In step S506, the system control unit **109** determines that the tally signal receiving unit **122** is in the state “tally signal: PROGRAM” and controls the display unit **121** in such a way that the display unit **121** does not display the return video but displays the self-shot video. The process then returns to step S500.

[0085] In step S507, the system control unit **109** controls the display unit **121** in such a way that the display unit **121** displays the self-shot video. The process then returns to step S500.

[0086] With the process shown in FIG. 5, as described above, the system control unit **109** switches a method for displaying a return video in accordance with the need to display the return video which depends on the state of the tally signal receiving unit **122**. Consequently, it is possible to produce an effect of reducing the risk of the display of the return video inhibiting the display of the self-shot video on the display unit **121**.

[0087] It should be noted that, in the present embodiment, the displayed size, displayed/non-displayed state, and transmittance of the return video are changed depending on the state of the tally signal receiving unit **122**; however, the present invention is not limited to this scheme. If it is possible to produce the same effect as the above, for example, the method for displaying the self-shot video may be switched depending on the state of the tally signal receiving unit **122**.

[0088] The present embodiment has been described regarding the case where the output destination to which the information about the method for displaying the return video is to be output is the display unit **121**; however, the output destination may be at least one of a plurality of display parts and an external recording apparatus. Moreover, the method for switching the display of the return video depending on the state of the tally signal receiving unit **122** may be set changeable through a user's operation on the operation unit **111** for each output destination.

Other Embodiments

[0089] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0090] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The

scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0091] This application claims the benefit of Japanese Patent Application No.2024-031326, filed Mar. 1, 2024 which is hereby incorporated by reference herein in its entirety.

Claims

1. An image pickup apparatus connected to an external apparatus, comprising: at least one memory and at least one processor which function as: a first receiving unit configured to receive a tally signal; a second receiving unit configured to receive a return video; and a control unit configured to switch a method for displaying the return video, wherein the control unit switches the method for displaying the return video, depending on a state of the tally signal in the first receiving unit.
2. The image pickup apparatus according to claim 1, wherein in a case where, even when the return video is set to be displayed simultaneously with a self-shot video being shot by the image pickup apparatus, the first receiving unit enters a first state where a tally signal of PROGRAM is being received as the tally signal, the control unit switches the method for displaying the return video to a method for displaying only the self-shot video by setting the return video to a non-displayed state.
3. The image pickup apparatus according to claim 2, wherein in a case where the return video is set to be displayed simultaneously with the self-shot video, and the first receiving unit enters a second state where the tally signal is not being received, the control unit switches the method for displaying the return video to a method for displaying the return video simultaneously with the self-shot video.
4. The image pickup apparatus according to claim 3, wherein in a case where the return video is set to be displayed simultaneously with the self-shot video, and the first receiving unit enters a third state where a tally signal of PREVIEW is being received as the tally signal, the control unit switches the method for displaying the return video in such a way that the return video is displayed simultaneously with the self-shot video, with visibility of the self-shot video being higher than the visibility of the self-shot video in the first state.
5. The image pickup apparatus according to claim 2, wherein in a case where, even when the return video is set to be displayed in a PinP mode with respect to the self-shot video, the first receiving unit enters the first state, the control unit switches the method for displaying the return video to a method for displaying only the self-shot video by setting the return video to a non-displayed state.
6. The image pickup apparatus according to claim 3, wherein in a case where the return video is set to be displayed in a PinP mode with respect to the self-shot video, and the first receiving unit enters the second state, the control unit switches the method for displaying the return video to a method for displaying the return video in the PinP mode.
7. The image pickup apparatus according to claim 4, wherein in a case where the return video is set to be displayed in a PinP mode with respect to the self-shot video, and the first receiving unit enters the third state, the control unit switches the method for displaying the return video to a method for displaying the return video in the PinP mode with a displayed size of the return video decreased.
8. The image pickup apparatus according to claim 4, wherein in a case where the return video is set to be displayed in a PinP mode with respect to the self-shot video, and the first receiving unit enters the third state, the control unit switches the method for displaying the return video to a method for displaying the return video in the PinP mode with transmittance of the return video increased.
9. The image pickup apparatus according to claim 2, wherein in a case where, even when the return video is set to be displayed in a mixing mode with respect to the self-shot video, the first receiving unit enters the first state, the control unit switches the method for displaying the return video to a method for displaying only the self-shot video by setting the return video to a non-displayed state.
10. The image pickup apparatus according to claim 3, wherein in a case where the return video is set to be displayed in a mixing mode with respect to the self-shot video, and the first receiving unit

enters the second state, the control unit switches the method for displaying the return video to a method for displaying the return video in the mixing mode.

11. The image pickup apparatus according to claim 4, wherein in a case where the return video is set to be displayed in a mixing mode with respect to the self-shot video, and the first receiving unit enters the third state, the control unit switches the method for displaying the return video to a method for displaying the return video in the mixing mode with transmittance of the return video increased.

12. The image pickup apparatus according to claim 2, wherein in a case where, even when the return video is set to be displayed in a screen separation mode with respect to the self-shot video, the first receiving unit enters the first state, the control unit switches the method for displaying the return video to a method for displaying only the self-shot video by setting the return video to a non-displayed state.

13. The image pickup apparatus according to claim 3, wherein in a case where the return video is set to be displayed in a screen separation mode with respect to the self-shot video, and the first receiving unit enters the second state, the control unit switches the method for displaying the return video to a method for displaying the return video in the screen separation mode.

14. The image pickup apparatus according to claim 4, wherein in a case where the return video is set to be displayed in a screen separation mode with respect to the self-shot video, and the first receiving unit enters the third state, the control unit switches the method for displaying the return video to a method for displaying the return video in the screen separation mode with a displayed size of the return video decreased but a displayed size of the self-shot video increased.

15. The image pickup apparatus according to claim 2, wherein the setting in which the return video is displayed simultaneously with the self-shot video is set changeable, in response to a user's operation, to one of a first setting in which the return video is displayed in a PinP mode with respect to the self-shot video, a second setting in which the return video is displayed in a mixing mode with respect to the self-shot video, and a third setting in which the return video is displayed in a screen separation mode with respect to the self-shot video.

16. The image pickup apparatus according to claim 7, wherein the displayed size of the return video in the PinP mode in each of the second state and the third state is set changeable through a user's operation.

17. The image pickup apparatus according to claim 8, wherein the transmittance of the return video displayed in the PinP mode in each of the second state and the third state is set changeable in response to a user's operation.

18. The image pickup apparatus according to claim 11, wherein the transmittance of the return video displayed in the mixing mode in each of the second state and the third state is set changeable in response to a user's operation.

19. The image pickup apparatus according to claim 14, wherein the displayed sizes of the return video and the self-shot video in each of the second state and the third state is set changeable through a user's operation.

20. The image pickup apparatus according to claim 1, wherein the method for displaying the return video depending on the state of the first receiving unit is set changeable for each output destination to which the control unit outputs the information about the method for displaying the return video.

21. The image pickup apparatus according to claim 20, wherein the output destination is at least one of a display unit and an external recording apparatus.

22. A control method for an image pickup apparatus connected to an external apparatus, wherein the method comprises: a first receiving step of receiving a tally signal; a second receiving step of receiving a return video; and a control step of switching a method for displaying the return video with respect to a self-shot video, wherein in the control step, the method for displaying the return video is switched depending on a state of the tally signal in the first receiving step.

23. A non-transitory computer readable storage medium storing a program that causes a computer

to execute a control method for an image pickup apparatus connected to an external apparatus, wherein the method comprises: a first receiving step of receiving a tally signal; a second receiving step of receiving a return video; and a control step of switching a method for displaying the return video with respect to a self-shot video, wherein in the control step, the method for displaying the return video is switched depending on a state of the tally signal in the first receiving step.
